Modelling a De Laval nozzle

CFD study under STAR-CCM+

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IPSA — March 2026

Report reformatted in October 2026, content and results unchanged

Abstract. Compressible flow through a convergent-divergent nozzle is simulated in STAR-CCM+ for two back pressures (300 kPa and 500 kPa) and compared with isentropic theory: exit pressure, mass flow rate, and the evolution of pressure, temperature and Mach number along the nozzle, including the position of the normal shock.

Introduction

This report presents a Computational Fluid Dynamics (CFD) study of a De Laval (convergent-divergent) nozzle, carried out using Simcenter STAR-CCM+. The De Laval nozzle is a fundamental component in aerospace propulsion systems, designed to accelerate a subsonic flow to supersonic velocities by exploiting the compressible nature of gases at high Mach numbers.

The objective of this study is to simulate and analyse the internal flow behaviour of the nozzle under two distinct back-pressure conditions — 300 kPa and 500 kPa — and to compare the numerical results against analytical predictions derived from isentropic flow theory. Key quantities of interest include the exit static pressure, the mass flow rate, and the spatial evolution of pressure, temperature and Mach number along the nozzle axis.

The report is structured as follows: an analytical study first establishes the theoretical reference values, followed by a mesh convergence analysis to ensure numerical accuracy. The physical models selected in STAR-CCM+ are then justified, and the simulation results for both pressure conditions are presented and discussed.

1 Analytical study

1.1 Exit pressure

In order to determine the theoretical static pressure at the nozzle exit, the isentropic flow relation between static and total pressure is applied. For a perfect gas flowing isentropically through a convergent-divergent nozzle, this relation is expressed as

$$\frac{P_s}{P_0} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}}. \tag{1}$$

The exit static pressure is therefore

$$P_s = P_0 \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\frac{\gamma}{\gamma-1}} = 1\,000\,000 \left(1 + \frac{1.4 - 1}{2} \times 2.94^2\right)^{-\frac{1.4}{1.4-1}} = 29\,795\,\text{Pa}. \tag{2}$$

This result is consistent with the highly expanded supersonic regime expected at the nozzle exit: the static pressure is roughly 34 times lower than the stagnation pressure, reflecting the significant kinetic energy gained by the flow as it accelerates to $M = 2.94$.

1.2 Mass flow rate

Under choked conditions, the mass flow rate depends solely on the stagnation state and the throat area, and is given by

$$\dot{m} = P_0 A_c \sqrt{\frac{k}{RT_0}\left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}}}. \tag{3}$$

Applying this formula to the nozzle geometry gives $\dot{m} = 3.61\,\mathrm{kg/s}$.

This value will serve as the reference against which the CFD-predicted mass flow rate is compared in the subsequent sections.

2 Mesh convergence study

Prior to running the main simulations, a mesh convergence study was conducted to determine the optimal mesh resolution. The goal is to identify the coarsest mesh that still produces mesh-independent results, ensuring a good balance between numerical accuracy and computational efficiency.

Five different base mesh sizes were tested, ranging from 5 mm down to 0.1 mm. For each mesh, the absolute pressure at the nozzle outlet was recorded as the convergence indicator. The results are summarised in the table and chart below.

Mesh	Base size (mm)	Outlet pressure (Pa)
1	5	65 453
2	1	95 612
3	0.6	94 578
4	0.4	Error
5	0.1	Error

Table 1. Mesh convergence study.

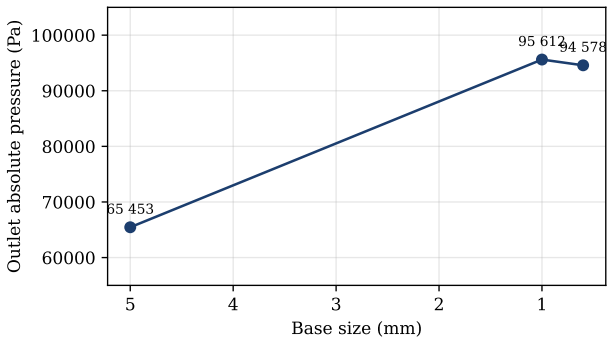


Figure 1. Absolute pressure at the outlet versus base size.

The analysis reveals three distinct behaviours. Mesh 1 (base size 5 mm) is clearly too coarse, yielding an outlet pressure of only 65 453 Pa — significantly lower than the values obtained with finer meshes, indicating that the shock structure and boundary layer are insufficiently resolved. Meshes 2 and 3 (base sizes 1 mm and 0.6 mm respectively) produce very close results — 95 612 Pa and 94 578 Pa — a difference of less than 1.1 %, which confirms that the solution has reached mesh independence beyond this level of refinement. Meshes 4 and 5 (base sizes 0.4 mm and 0.1 mm) both generated solver errors, most likely due to an excessive cell count exceeding the available computational resources.

Based on these results, **mesh 3 with a base size of** 0.6 mm was selected for all subsequent simulations. It provides a mesh-independent solution while remaining computationally feasible.

3 Final mesh

The figure below shows the final mesh used for all simulations, generated with a base size of 0.6 mm as determined by the convergence study. The mesh is composed of polyhedral cells, which are well suited to the curved geometry of the convergent-divergent nozzle walls, as they adapt naturally to complex boundaries while maintaining a good aspect ratio.

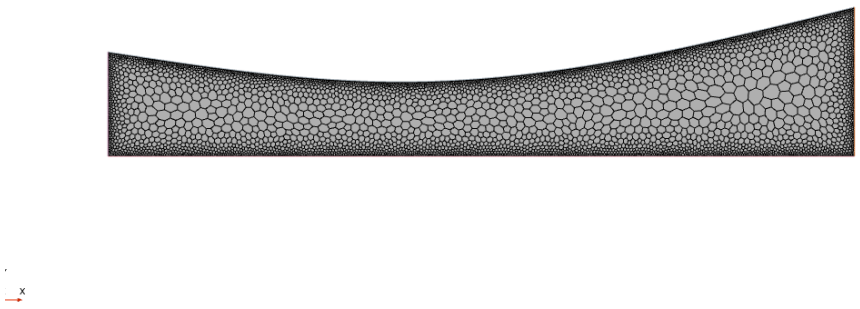


Figure 2. Final polyhedral mesh (base size 0.6 mm).

As visible in the figure, the cell density is higher in the throat region and along the nozzle axis, where the steepest gradients in velocity, pressure and temperature are expected. The cells progressively grow in size toward the inlet and outlet boundaries, where the flow is more uniform and requires less resolution.

The simulation is performed on a 2D axisymmetric domain, which exploits the rotational symmetry of the nozzle geometry about its longitudinal axis. This approach significantly reduces the total cell count compared to a full 3D mesh, while retaining complete physical accuracy for axisymmetric flow conditions.

4 Physics justification

The physical models activated in STAR-CCM+ were selected to accurately represent the compressible, turbulent flow regime expected inside the Laval nozzle. The rationale for each key model is detailed below.

Axisymmetric. The nozzle geometry is rotationally symmetric about its longitudinal axis. Activating this model allows the full 3D flow to be represented on a 2D meridional plane, significantly reducing computational cost without any loss of physical accuracy for this type of geometry.

Ideal gas. Air is modelled as an ideal gas, which is a standard and appropriate assumption for the pressure and temperature conditions of this study. It provides the equation of state $P = \rho RT$ that closes the compressible flow equations.

Coupled flow & coupled energy. The coupled solver is used instead of the segregated solver because it solves the continuity, momentum and energy equations simultaneously. This is essential for high-speed compressible flows where strong pressure–velocity–energy coupling exists, particularly in the transonic and supersonic regime.

SST (Menter) k–ω. The Shear Stress Transport k–ω model is selected for turbulence closure. It combines the robustness of the k–ω formulation near walls with the free-stream independence of the k–ε model in the core flow, making it particularly well suited for flows with adverse pressure

gradients and shock–boundary layer interactions, which are expected in the diverging section of the nozzle.

All y^+ wall treatment. This blended wall treatment remains valid over a wide range of near-wall mesh resolutions, ensuring that wall shear stress is correctly computed regardless of whether the first cell falls within the viscous sublayer or the log-law region.

Steady. A steady-state solver is used since the boundary conditions are time-invariant. This assumption is valid as long as the flow does not exhibit strong unsteady phenomena such as shock oscillations, which was verified through residual monitoring during the simulations.

5 Study

5.1 Back pressure of 300 kPa

In this first case, a back pressure of 300 kPa is imposed at the nozzle exit. Since this value is significantly higher than the design exit pressure of 29 795 Pa, the nozzle operates in an off-design, over-expanded regime.

Flow behaviour

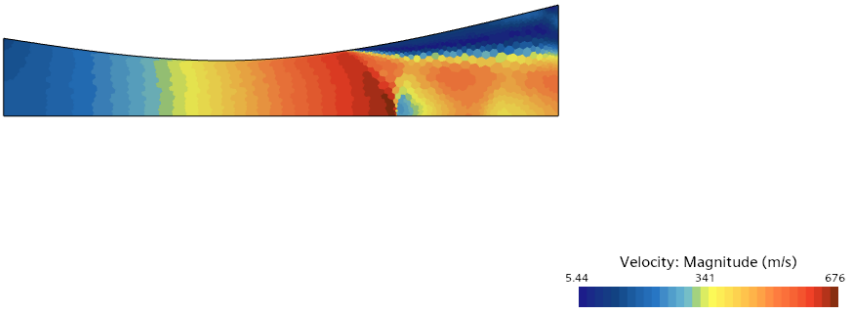


Figure 3. Velocity magnitude, back pressure 300 kPa.

The velocity contour plot shows the flow accelerating from subsonic conditions at the inlet to supersonic velocities through the diverging section, reaching a peak of approximately 676 m/s. A sharp velocity discontinuity is clearly visible midway through the diverging section, indicating the presence of a normal shock wave. This shock is a direct consequence of the imposed back pressure of 300 kPa being too high for the nozzle to operate under fully expanded supersonic conditions.

Pressure profile

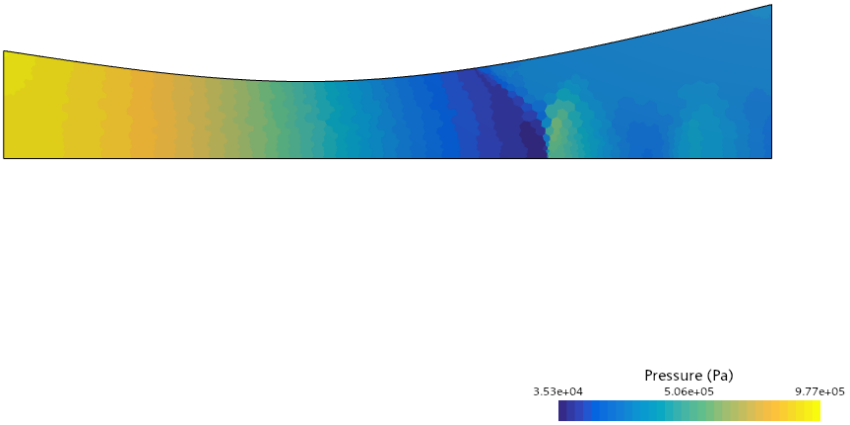


Figure 4. Pressure field, back pressure 300 kPa.

The pressure contour map shows a continuous decrease from approximately 977 000 Pa at the inlet down to around 35 300 Pa in the supersonic region of the diverging section. A sharp pressure recovery is then visible at the shock location, where the pressure rises abruptly back toward the imposed back pressure of 300 kPa. This pressure jump is the hallmark signature of a normal shock wave, confirming the observations made in the velocity field.

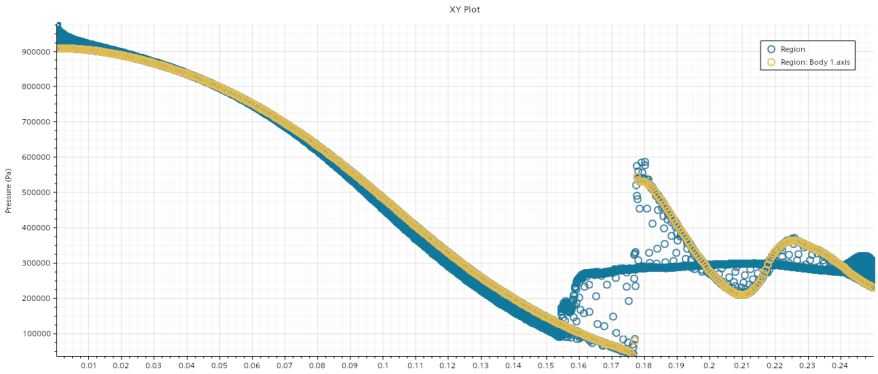


Figure 5. Pressure along the nozzle axis and wall, back pressure 300 kPa.

The XY plot confirms the pressure evolution along the nozzle axis. The pressure decreases continuously and smoothly from $\sim 950\,000$ Pa at the inlet down to a minimum of $\sim 100\,000$ Pa just before the shock, located at approximately $x = 0.17$ m. At this point, the pressure rises sharply across the normal shock before oscillating and stabilising around the imposed back pressure of 300 000 Pa at the exit. The close overlap between the axis and wall profiles upstream of the shock confirms the predominantly one-dimensional character of the flow in that region, while the scatter downstream reflects the complex post-shock behaviour.

Temperature and Mach number profiles

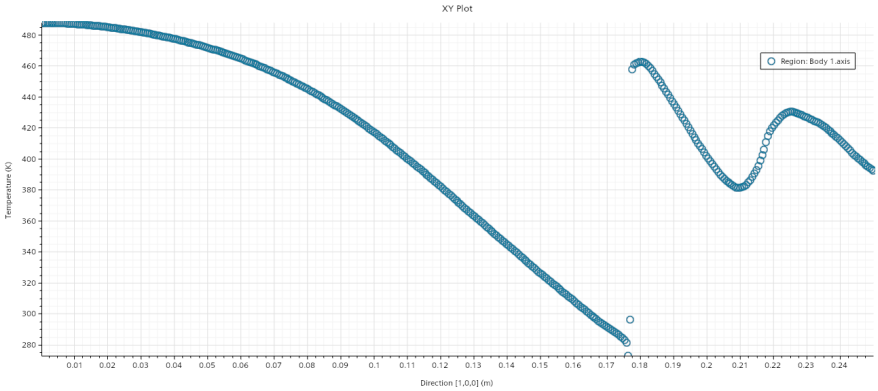


Figure 6. Temperature along the nozzle, back pressure 300 kPa.

The temperature profile shows a continuous decrease from approximately 480 K at the inlet down to around 280 K just upstream of the shock, consistent with the isentropic expansion of the flow as it accelerates through the diverging section. Across the normal shock at $x \approx 0.17$ m, the temperature rises sharply as kinetic energy is partially converted back into thermal energy through the irreversible shock process. Downstream, the temperature stabilises at around 380 K to 430 K toward the nozzle exit.

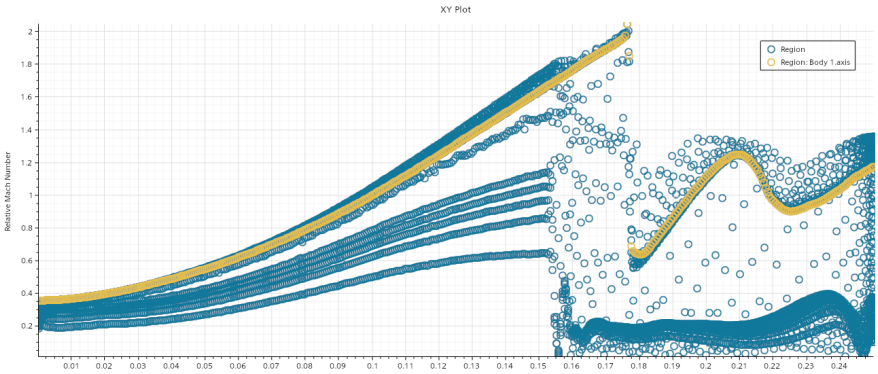


Figure 7. Mach number along the nozzle, back pressure 300 kPa.

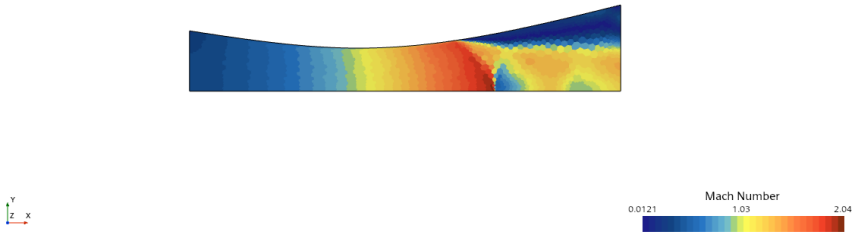


Figure 8. Mach number field, back pressure 300 kPa.

The Mach number profile confirms that the flow accelerates continuously from near zero at the inlet, reaching $M = 1$ at the throat before continuing to increase supersonically up to approximately $M = 2$ in the diverging section. At $x \approx 0.17$ m, the Mach number drops abruptly across the normal shock to subsonic values around $M \approx 0.5$, after which it remains subsonic through to the exit. The contour map clearly illustrates the supersonic core ahead of the shock and the

subsonic region downstream. The significant scatter in the XY plot past the shock reflects the three-dimensional post-shock flow structures captured across the full cross-section.

Comparison with analytical results

The analytical study predicted an exit static pressure of 29 795 Pa and a mass flow rate of 3.61 kg/s, both derived from isentropic flow theory under fully expanded supersonic conditions at $M = 2.94$. The CFD simulation under a 300 kPa back pressure does not reach these design conditions, as the normal shock prevents the flow from fully expanding to the design Mach number. The peak Mach number observed before the shock is approximately $M = 2.0$, lower than the design value of 2.94, and the exit pressure is governed by the imposed back pressure rather than the isentropic design value. However, the mass flow rate remains in good agreement with the analytical prediction, confirming that the throat is correctly choked in the simulation. The small discrepancy is attributable to viscous and turbulent effects not captured by the inviscid analytical model.

5.2 Back pressure of 500 kPa

Flow behaviour

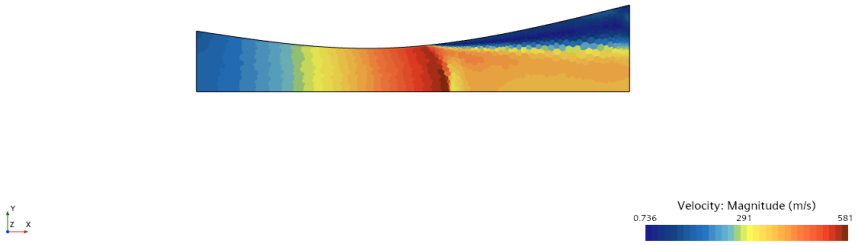


Figure 9. Velocity magnitude, back pressure 500 kPa.

Compared to the 300 kPa case, the higher back pressure pushes the normal shock further upstream in the diverging section. The peak velocity before the shock is approximately 581 m/s, lower than the 676 m/s observed previously, reflecting the reduced Mach number achieved before the shock occurs. The supersonic core is noticeably shorter, and the subsonic deceleration region occupies a larger portion of the diverging section.

Pressure profile

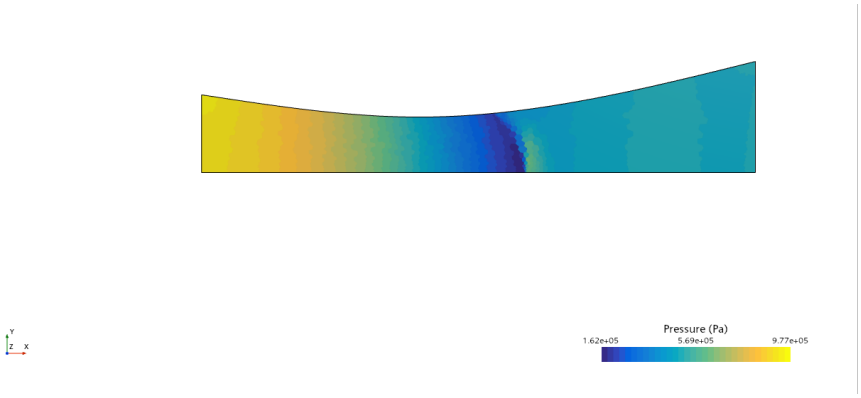


Figure 10. Pressure field, back pressure 500 kPa.

The pressure contour map shows a similar trend to the 300 kPa case, with a continuous decrease from approximately 977 000 Pa at the inlet toward the throat. However, the pressure recovery across the shock occurs earlier in the diverging section, consistent with the upstream shift of the normal shock. The minimum pressure reached before the shock is higher than in the 300 kPa case, reflecting the lower Mach number achieved. Downstream of the shock, the pressure recovers smoothly toward the imposed back pressure of 500 kPa.

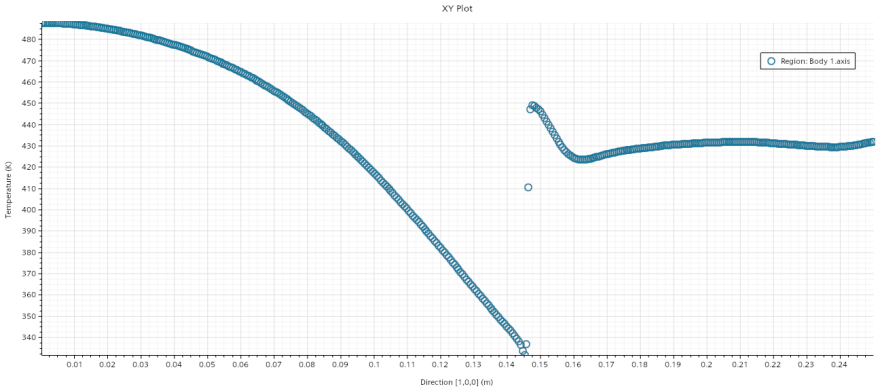


Figure 11. Temperature along the nozzle, back pressure 500 kPa.

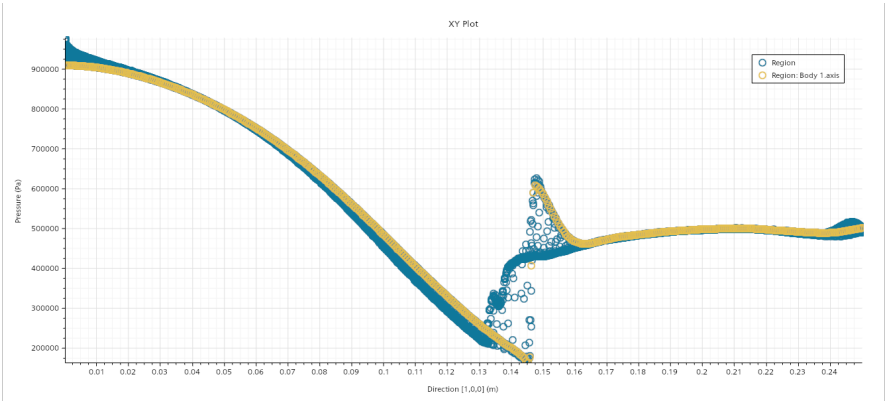


Figure 12. Pressure along the nozzle axis and wall, back pressure 500 kPa.

The XY plot confirms the upstream shift of the normal shock, now located at approximately $x = 0.14$ m compared to $x = 0.17$ m in the 300 kPa case. The pressure drops from $\sim 950\,000$ Pa at the inlet to a minimum of $\sim 200\,000$ Pa before rising sharply across the shock. The post-shock pressure recovery is smoother and less oscillatory than in the 300 kPa case, which is consistent with a weaker shock occurring at a lower Mach number. The pressure then stabilises around the imposed back pressure of 500 000 Pa toward the exit.

Temperature and Mach number profiles

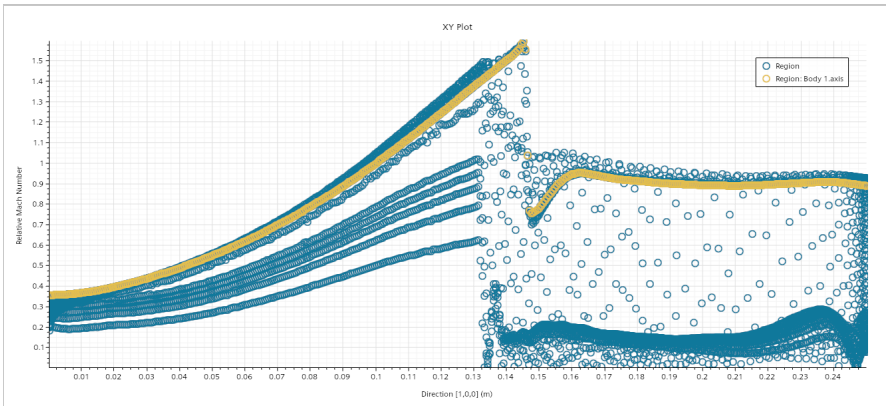


Figure 13. Mach number along the nozzle, back pressure 500 kPa.

The Mach number profile confirms that the peak value reached before the shock is approximately $M = 1.59$, significantly lower than the $M = 2.0$ observed in the 300 kPa case, consistent with the earlier shock location. The flow accelerates from $M = 0$ at the inlet to $M = 1$ at the throat before continuing to increase supersonically up to the shock at $x \approx 0.14$ m, where it drops abruptly to subsonic values. The scatter downstream of the shock is less pronounced than in the 300 kPa case, reflecting the weaker post-shock disturbances associated with a lower upstream Mach number.

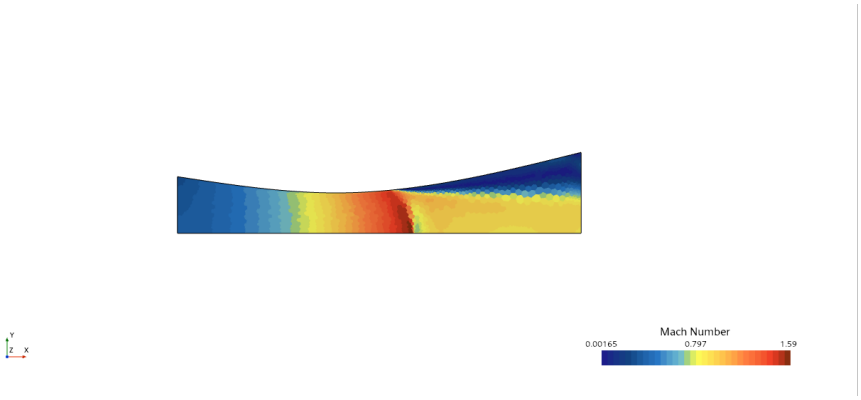


Figure 14. Mach number field, back pressure 500 kPa.

Comparison with analytical results

As in the 300 kPa case, the 500 kPa back pressure prevents the nozzle from reaching its design conditions. The normal shock occurs earlier, at $x \approx 0.14$ m, limiting the peak Mach number to approximately 1.59, further from the analytical design value of 2.94. The exit pressure is once again governed by the imposed back pressure rather than the isentropic design value of 29 795 Pa.

The mass flow rate, however, remains consistent with the analytical prediction of 3.61 kg/s, confirming that the throat remains correctly choked regardless of the back pressure imposed at the outlet. This is expected as long as the back pressure does not exceed the critical value required to unchoke the throat.

Comparing both cases, the key difference lies in the shock position and intensity: the higher the back pressure, the further upstream the shock migrates and the weaker its intensity, which is fully consistent with classical convergent-divergent nozzle theory.

Conclusion

This study successfully demonstrated the capability of STAR-CCM+ to simulate compressible flow through a De Laval nozzle under two distinct off-design back-pressure conditions.

In both cases, the throat correctly reaches sonic conditions and the simulated mass flow rate is in good agreement with the analytical prediction of 3.61 kg/s, validating the mesh set-up and the physical model selection. The pressure, temperature and Mach number profiles all exhibit the expected isentropic behaviour upstream of the shock, while the shock-jump relations are well captured across the normal shock.

The comparison between the two cases highlights a fundamental principle of convergent-divergent nozzle aerodynamics: as the back pressure increases, the normal shock migrates upstream toward the throat, reducing the peak Mach number achieved before the shock. At 300 kPa, the shock occurs at $x \approx 0.17$ m with a peak Mach number of approximately 2.0, while at 500 kPa it shifts to $x \approx 0.14$ m with a peak of approximately 1.59.

Overall, the CFD results are physically consistent and in good qualitative agreement with analytical predictions, confirming the validity of the simulation approach adopted in this study.